

DIGIVOTE

Secure and Intelligent Online Voting Platform

AI Assistant Knowledge Base and Full Project Workflow

Purpose of this PDF

This document is written as a structured knowledge source for the DigiVote AI Assistant. It explains the project, its modules, technologies, security goals, and the end-to-end workflow so the assistant can answer project-related questions consistently.

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Source basis: First Review PPT, with the project's current implementation workflow and AI-assistant safety boundaries clearly identified where applicable.

1. Knowledge Base Purpose and Scope

DigiVote is a secure, web-based digital voting platform intended to simplify and modernize the traditional voting process. The first-review project description identifies secure authentication, identity verification, electronic vote casting, election management, result/report generation, and an AI chatbot as the core areas of the system.

This PDF is designed for retrieval-augmented generation (RAG). Important project facts are repeated in clear language so an AI assistant can answer questions about what DigiVote is, how each module works, what technologies are used, and how the modules connect.

Answering principle

When a question is about DigiVote, the assistant should prefer the project facts in this document over generic assumptions. When this document does not specify an exact implementation value (for example, a timeout duration), the assistant should say that the value is configurable or not specified rather than inventing a number.

Document map

- Project overview and objectives
- System architecture and technology stack
- Authentication modules and workflow
- Election Creation module
- Election Management module
- Voter Verification module
- Voting module
- Result and Report module
- AI Chatbot module
- End-to-end DigiVote workflow
- Security and privacy principles
- Hardware and software requirements
- Common questions and answer-ready facts
- AI assistant response boundaries

2. Project Overview

DigiVote (Digital Voting System with AI Chatbot) is a secure, web-based voting platform. It is designed to authenticate users, verify eligibility and identity through configured validation mechanisms, authorize one-time voting, store ballots confidentially, count votes automatically, publish results, and assist users with election-related questions.

2.1 Core modules

Module	What it does
Authentication Module	Secure registration, login, email verification, Google OAuth, OTP-based MFA, session management, and security controls.
Election Creation Module	Election details/type configuration, voter and candidate configuration, election-specific verification setup, and schedule/rules configuration.
Voter Verification Module	Voter eligibility validation, email OTP verification, webcam/biometric verification, and one-time voting authorization.
Election Management Module	Voter/candidate management, election start/stop control, configuration lock after start, monitoring, and audit logs.
Voting Module	Secure voting authorization, candidate selection, vote confirmation, one-time submission, and confidential ballot storage.
Result & Report Module	Automatic counting, result generation and publication, participation reports, and candidate/election reports.
AI Chatbot Module	Election-related question answering, voter/election-creator assistance, context-aware support, and knowledge-base/unanswered-question management.

2.2 Main objectives

- Modernize the traditional voting process through a secure web platform.
- Provide multiple authentication and verification mechanisms.
- Restrict voting to eligible voters and enforce one-time voting authorization.
- Give election creators tools to configure and manage elections.
- Provide secure vote submission and confidential ballot storage.
- Automate vote counting and result/report generation.
- Provide an AI assistant for project and election-related guidance.

3. System Architecture and Technology Stack

3.1 High-level architecture

The system is organized as a web application. The frontend provides user interfaces, the Django backend exposes application logic and APIs, PostgreSQL stores application data, authentication and verification services protect access, and the AI chatbot uses a knowledge base to answer election-related questions.

Logical flow: User -> React/Vite interface -> Django/DRF backend -> PostgreSQL and application services -> response to frontend. For AI assistance, the backend uses the knowledge base/RAG pipeline and Gemini/Pinecone services configured for the AI Assistant.

AI retrieval flow

Knowledge document -> text extraction -> chunking -> embeddings -> Pinecone retrieval -> relevant context -> Gemini answer. The AI assistant should use the backend for authorized election-specific context rather than directly exposing database contents.

3.2 Technology stack from the project

Layer	Technology / requirement
Frontend	React.js + Vite
Backend	Python 3.11 + Django + Django REST Framework
Database	PostgreSQL
AI / Face Verification	OpenCV + Face Recognition / ArcFace
Authentication	Google OAuth + OTP
Development Tools	VS Code + Git
API Testing	Postman

3.3 Hardware requirements

Component	Minimum requirement
Processor	Intel Core i3
RAM	8 GB
Storage	256 GB SSD
Camera	Built-in Webcam
Display	1366 x 768 resolution
Internet	Stable Internet Connection
GPU	Not required

3.4 Software requirements

Software / Technology	Requirement
Operating System	Windows 11 (64-bit)
Frontend	React.js + Vite
Backend	Python 3.11 + Django + Django REST Framework
Database	PostgreSQL
AI / Face Verification	OpenCV + Face Recognition / ArcFace
Authentication	Google OAuth + OTP
Development Tools	VS Code + Git
API Testing	Postman

4. Authentication Module

The DigiVote first-review design describes seven core authentication/security modules that work together to provide registration, multi-factor authentication, session management, and protection against unauthorized access.

4.1 Register Module

Purpose: allow a new user to create a DigiVote account securely with validation before account activation.

- Collects name, email, mobile number, and password.
- Enforces strong password validation at input.
- Prevents duplicate registrations.
- Places a newly registered account into Pending Email Verification status.

4.2 Login Module

Purpose: authenticate an existing user and verify account status before full access is granted.

- Uses email and password as the primary authentication factor.
- Checks account status before granting access.
- Provides informative invalid-login handling.
- Monitors login attempts for suspicious behavior.
- After successful password authentication, proceeds to OTP/MFA.

4.3 Google Authentication

Purpose: allow users to create or access a DigiVote account through Google OAuth.

- Uses the Google OAuth authentication flow.
- Validates the Google token on the backend.
- Can link an existing DigiVote account.
- Verifies Google account identity.
- Does not store the user's Google password.

4.4 Email Verification

Purpose: confirm that the email address supplied during registration belongs to the registering user.

- Generates a verification email and token after registration.
- Uses a time-limited verification link.
- Uses a single-use token to prevent replay.
- Supports resend with a cooldown period.
- Updates the email-verified status after successful confirmation.

4.5 OTP Authentication

Purpose: add a second authentication factor after password authentication.

- Generates a cryptographically secure 6-digit OTP per session.
- Applies an expiry time and maximum attempts.
- Stores the OTP as a hash rather than persisting plaintext.
- Uses resend cooldown and rate limiting to reduce abuse.
- Creates the authenticated session only after correct OTP entry.

4.6 Session Management

Purpose: securely create, monitor, and revoke active login sessions across devices.

- Creates secure sessions after successful OTP verification.
- Lets users view/manage active sessions.
- Automatically expires sessions after a defined inactivity period.
- Supports current-device logout and full active-session revocation.

4.7 Security Module

Purpose: protect authentication from brute-force attacks, unauthorized access, and session-based attacks.

- Login rate limiting.
- Failed login monitoring and suspicious-activity flagging.
- Account lock or suspension for suspicious behavior.
- Strong cryptographic password hashing.
- Comprehensive audit logging of authentication events and suspicious activity.

5. Authentication Workflow

The first-review PPT shows authentication as the entry point for access validation and depicts registration, login, Google authentication, OTP/MFA, and secure session establishment as the access-control path.

1. New user path: Register -> enter name/email/mobile/password -> validation -> Pending Email Verification.
2. Email confirmation: verification email/token -> user follows the time-limited link -> single-use token is validated -> email-verified state is updated.
3. Login path: enter email/password -> account status and credentials are checked -> login attempt monitoring applies.
4. Second factor: user proceeds to OTP/MFA -> enters the one-time code -> attempt/expiry/rate limits are enforced.
5. Session creation: correct OTP -> secure authenticated session is created.
6. Active use: session is monitored; users can view/revoke active sessions and log out.
7. Security enforcement runs across the flow through rate limiting, failed-login monitoring, account protection, hashing, and audit logging.

Google path

Google OAuth provides an alternate sign-in/registration path. The backend validates the Google token, verifies identity, and can link the Google identity to an existing DigiVote account. The user's Google password is never stored by DigiVote.

6. Election Creation Module

The Election Creation module defines the configuration of an election before voting begins. The first-review PPT identifies four main areas: election details/type, voter and candidate configuration, election-specific verification setup, and schedule/rules configuration.

6.1 Election configuration

- Define election details and election type.
- Configure the voter population and candidate information.
- Set verification mechanisms for the specific election.
- Configure election schedule and rules.

6.2 Conceptual creator workflow

8. Authenticated user opens the election creation area.
9. Enters election details and selects the election type/configuration.
10. Configures candidates and eligible voters for the election.
11. Selects the verification mechanisms that apply to this election.
12. Defines election schedule and rules.
13. Reviews the configuration and makes the election available according to the configured schedule.

Eligibility principle

DigiVote treats voter eligibility as election-specific in the current project workflow: a voter can participate in an election only when that voter is included in the eligible voter set configured for that election.

6.3 Current unified-account workflow

The current project workflow uses one DigiVote account rather than separate voter and election-creator accounts. The same account may create/manage an election it owns and may vote in a different election when included in that election's eligible voter list. The backend enforces access and ownership.

7. Election Management Module

Election Management controls the operational state of an election after it has been configured. The first-review PPT identifies voter/candidate management, election start/stop control, configuration lock after start, and monitoring/audit logs.

- Manage voters and candidates associated with the election.
- Start or stop an election according to its operational lifecycle.
- Lock election configuration after the election has started so core settings are not changed during voting.
- Monitor election activity and maintain audit logs.

7.1 Conceptual lifecycle

A practical end-to-end lifecycle for the current DigiVote workflow is: Draft configuration -> Scheduled (when a future start is defined) -> Live/active voting -> Completed. A paused/controlled state may be used by the application when supported by configuration. The exact status names are implementation-specific unless defined by the deployed application.

7.2 Why configuration locking matters

Once voting begins, changing core election configuration can affect fairness and auditability. The project therefore specifies configuration lock after start. Operational changes should be controlled and reflected in audit logs.

8. Voter Verification Module

The Voter Verification module determines whether a user can participate in a selected election and completes the required verification steps configured for that election.

- Voter eligibility validation.
- Email OTP verification.
- Webcam/biometric verification.
- One-time voting authorization.

8.1 Conceptual voter verification workflow

14. User selects a particular election after signing in.
15. The system checks whether the user is present in that election's eligible voter set.
16. If eligible, the system applies the election's configured verification mechanisms, such as email OTP and/or webcam/biometric verification.
17. After successful verification, the system issues one-time voting authorization for that election.
18. The user is allowed to proceed to the Voting module only when the authorization is valid.

Identity boundary

Webcam/biometric verification is an identity-validation mechanism. Face-verification technology in the stated stack includes OpenCV with Face Recognition / ArcFace. The first-review PPT does not specify a particular face-matching algorithm configuration or enrollment workflow, so the assistant should not invent one.

9. Voting Module

The Voting module handles the actual ballot-selection and submission process after the voter has been authenticated, verified, and authorized.

- Secure voting authorization.
- Candidate selection and vote confirmation.
- One-time vote submission.
- Confidential ballot storage.

9.1 Conceptual voting workflow

19. User enters the selected election through the authorized voting path.
20. System confirms that the user has valid one-time voting authorization.
21. Voting interface displays the candidates available for the election.
22. Voter selects a candidate and reviews the selection.
23. Voter confirms the vote.
24. System accepts the vote as a one-time submission and stores the ballot confidentially.
25. The voter cannot use the same one-time authorization again.

Confidentiality

The project specifies confidential ballot storage. The AI Assistant must not expose private ballot contents or reveal another person's vote choice.

10. Result and Report Module

The Result & Report module turns submitted votes into election outcomes and reporting information.

- Automatic vote counting.
- Result generation and publishing.
- Voter participation reports.
- Candidate and election reports.

10.1 Conceptual result workflow

26. Voting closes according to the election schedule or management action.
27. The system performs automatic vote counting over the stored ballots.
28. Election results are generated.
29. Results are published according to the system workflow.
30. Participation and candidate/election reports are generated for appropriate users.

AI boundary

The assistant can explain published result information and reporting concepts, but it must not reveal confidential ballots or private vote choices.

11. AI Chatbot / AI Assistant Module

The project includes an AI chatbot to provide election-related question answering, assistance to voters and election creators, context-aware election support, and knowledge-base/unanswered-question management.

11.1 What the AI assistant is for

- Explain DigiVote concepts and modules.
- Answer questions about authentication, verification, voting, elections, and reports.
- Provide user assistance based on the project knowledge base.
- Use authorized election context when the application supplies it.
- Retrieve relevant information from the knowledge base before generating an answer.

11.2 RAG knowledge workflow

31. Project knowledge documents are converted into searchable text.
32. Text is divided into meaningful chunks with source/page metadata.
33. Gemini embeddings are generated for the chunks.
34. Embeddings are stored in Pinecone.
35. When a user asks a question, the question is embedded.
36. Pinecone retrieves the most relevant knowledge chunks.
37. Relevant chunks are combined into the context supplied to Gemini.
38. Gemini generates the answer using the retrieved project context and any authorized election context.

11.3 Context-aware assistance

For a question such as "Am I eligible to vote in this election?", the answer should not be guessed by the language model. The backend should use the authenticated user and the selected election to check the actual election-specific eligibility information, then use the knowledge base to explain the meaning of the result.

11.4 Knowledge-base management

The project scope includes knowledge-base support and unanswered-question management. This means the AI component can be maintained as a project knowledge source rather than being treated as a generic chatbot with no grounding.

12. Full End-to-End DigiVote Workflow

The following workflow connects the project modules into one coherent user journey. It is a conceptual workflow derived from the module functions stated in the first-review PPT and the project's current integrated architecture.

12.1 Complete workflow

- 1. Access DigiVote:** User opens the web application and reaches the authentication entry point.
- 2. Create or access account:** New user registers, or an existing user logs in. Google OAuth is an alternate sign-in path.
- 3. Verify email:** A newly registered user completes time-limited, single-use email verification.
- 4. Complete MFA:** Password authentication is followed by 6-digit OTP verification when the OTP/MFA path is required.
- 5. Secure session:** A secure authenticated session is established and can be monitored or revoked.
- 6. Use the dashboard:** The authenticated user can access the parts of DigiVote allowed by their data and election ownership/eligibility.
- 7. Create an election:** An election creator configures election details/type, voters, candidates, verification settings, schedule, and rules.
- 8. Manage election:** The creator manages voters/candidates, controls start/stop, and monitors activity with audit logs. Configuration locks after start.
- 9. Select an election as a voter:** A logged-in user selects a particular election in which they want to participate.
- 10. Check eligibility:** The system validates the user against that election's eligible voter set.
- 11. Complete required verification:** The system applies the election-specific verification configuration, such as email OTP and/or webcam/biometric verification.
- 12. Receive one-time voting authorization:** Successful verification enables the voter to enter the voting process.
- 13. Cast the vote:** The voter selects a candidate, confirms the selection, submits once, and the ballot is stored confidentially.
- 14. Count and publish:** After voting closes, votes are counted automatically and results/reports are generated and published as permitted.
- 15. Get help from AI:** At relevant points, the DigiVote Assistant answers project/election questions using RAG and authorized context without performing voting or administrative actions.

12.2 One-line workflow for quick retrieval

Master flow

Register / Google -> Email verification -> Login -> OTP/MFA -> Secure session -> Election creation/management OR Available election selection -> Election-specific eligibility -> Configured verification -> One-time voting authorization -> Candidate selection -> Vote confirmation -> One-time submission -> Confidential ballot storage -> Automatic counting -> Result/report publication -> AI assistance throughout the process.

13. Security, Privacy, and Trust Principles

Security is a cross-cutting concern in DigiVote. The first-review PPT emphasizes authentication protection, multi-factor verification, session control, rate limiting, failed-login monitoring, account locking/suspension, strong password hashing, and audit logging.

13.1 Authentication security

- Use email verification to validate registration email ownership.
- Use OTP as a second authentication factor when the MFA path is required.
- Store OTP as a hash rather than plaintext.
- Apply expiry, attempt limits, and resend cooldowns.
- Monitor and restrict suspicious login behavior.
- Allow active sessions to be viewed and revoked.
- Protect passwords with strong cryptographic hashing.

13.2 Election and voting security

- Validate voter eligibility for the specific election.
- Use one-time voting authorization.
- Restrict vote submission to the authorized flow.
- Store ballots confidentially.
- Lock election configuration after voting starts.
- Maintain election monitoring and audit logs.

13.3 AI security boundaries

- The AI assistant is informational; it does not cast votes.
- It does not change election settings or voter eligibility.
- It does not reveal private voter records.
- It does not reveal confidential ballots or vote choices.
- It does not recommend candidates or persuade voters.
- It must not expose API keys, passwords, tokens, or other secrets.
- When database facts are required, the backend is authoritative.

13.4 Current scope note for the user interface

The current DigiVote UI scope is intentionally focused on the voting platform itself. There should be no voter-facing Aadhaar, DigiLocker, Voter ID linking, or biometric-enrollment setup screens.

Webcam/biometric capability is used only as part of the configured verification path when required; it is not a separate enrollment flow.

14. Common Questions and Answer-Ready Facts

Q: What is DigiVote?

A: DigiVote is a secure, web-based digital voting platform designed to simplify and modernize the traditional voting process. It supports secure authentication, voter verification, electronic voting, election management, result/report generation, and an AI chatbot.

Q: What technologies are used?

A: React.js + Vite for the frontend; Python 3.11 + Django + Django REST Framework for the backend; PostgreSQL for the database; OpenCV + Face Recognition / ArcFace for AI/face verification; Google OAuth + OTP for authentication; VS Code + Git for development; Postman for API testing.

Q: What are the seven authentication/security modules?

A: Register, Login, Google Authentication, Email Verification, OTP Authentication, Session Management, and Security.

Q: How does registration work?

A: The user provides name, email, mobile number, and password. Password strength is validated, duplicate accounts are prevented, and the account starts in Pending Email Verification status.

Q: How does login work?

A: The user enters email and password. The system checks account status and login behavior, then proceeds to OTP/MFA before establishing the authenticated session when that path is required.

Q: How does Google login work?

A: The system uses Google OAuth, validates the Google token on the backend, verifies the Google identity, and can link an existing DigiVote account. The Google password is never stored by DigiVote.

Q: What is email verification?

A: It confirms that the email address used during registration is genuine. A time-limited, single-use verification link/token is sent, with resend cooldown protection.

Q: What is OTP authentication?

A: OTP authentication is the second factor after password authentication. DigiVote specifies a cryptographically secure 6-digit OTP with expiry, attempt limits, hashed storage, and resend/rate controls.

Q: What is session management?

A: It securely creates, monitors, expires, and revokes active sessions across devices. Users can log out from the current device or revoke all active sessions.

Q: How is voter eligibility handled?

A: Eligibility is checked for the selected election. A user must be included in that election's eligible voter set to participate.

Q: How is identity verified for voting?

A: Depending on the election configuration, DigiVote can use email OTP verification and webcam/biometric verification. The stated technology stack includes OpenCV with Face Recognition / ArcFace.

Q: What happens before voting?

A: The user is authenticated, checked for election-specific eligibility, completes the configured verification, and receives one-time voting authorization.

Q: Can a voter vote twice?

A: The design specifies one-time voting authorization and one-time vote submission, so the same authorization cannot be reused for another vote.

Q: How are votes stored?

A: The Voting module specifies confidential ballot storage.

Q: How are results produced?

A: Votes are counted automatically, results are generated and published, and voter participation plus candidate/election reports can be produced.

Q: What does the AI Assistant do?

A: It answers election-related questions, assists voters and election creators, provides context-aware support, and uses a knowledge base to ground answers.

Q: Can the AI Assistant cast my vote?

A: No. The assistant is informational and must not cast or modify votes.

Q: Can the AI Assistant recommend a candidate?

A: No. It can explain available candidate information but must remain neutral and should not recommend or rank candidates.

Q: Can the AI Assistant show all voters?

A: No. Private voter information must not be exposed.

Q: Can the AI Assistant reveal vote choices?

A: No. Confidential ballots and individual vote choices must not be exposed.

Q: What hardware is required?

A: Minimum stated hardware is Intel Core i3, 8 GB RAM, 256 GB SSD, built-in webcam, 1366 x 768 display, stable internet, and no GPU requirement.

Q: Which operating system is specified?

A: Windows 11 (64-bit).

15. AI Assistant Response Policy for DigiVote

This section is intended to guide grounded answers from the knowledge base and to prevent the assistant from inventing project facts or performing actions that belong to the secure application workflow.

15.1 Answer what is known

- Prefer this document for project-specific terminology and workflow.
- Use exact project names such as Authentication Module, Election Creation Module, Voter Verification Module, Election Management Module, Voting Module, Result & Report Module, and AI Chatbot Module.
- Do not invent exact timeout values, database fields, algorithm parameters, or deployment settings that are not specified in the knowledge base.
- When a detail is implementation-specific and not stated here, say that it depends on the current configuration or deployed application.

15.2 Protect private information

- Never expose another voter's private profile details.
- Never expose secret tokens, API keys, passwords, or credentials.
- Never reveal confidential ballot contents or another person's vote choice.
- Use only the minimum authorized election context necessary to answer the user's question.

15.3 Keep the assistant neutral

- Do not recommend or rank candidates.
- Do not persuade a voter.
- Do not predict a winner as a recommendation.
- Explain processes and published information instead.

15.4 Keep the assistant informational

- Explain how registration, verification, voting, elections, and reports work.
- Do not cast a vote.
- Do not modify elections or candidates.
- Do not change voter eligibility.
- Do not bypass verification or security controls.

15.5 Database-aware answers

When the user asks a question about their current election-specific status, such as eligibility, verification state, voting status, or election availability, the application backend should determine the current fact from authorized database state. Gemini should explain that authorized fact rather than invent it.

16. Suggested Questions for the DigiVote AI Assistant

Project overview

- What is DigiVote?
- What problem does DigiVote solve?
- What are the main modules?
- What is the technology stack?

Authentication

- How do I register?
- Why do I need email verification?
- How does OTP authentication work?
- How does Google authentication work?
- What is session management?
- How does DigiVote protect login attempts?

Election creation and management

- How is an election created?
- What can an election creator configure?
- How are voters and candidates configured?
- What is election-specific verification?
- What happens when an election starts?
- Why is election configuration locked after start?

Voter verification and voting

- How is voter eligibility checked?
- What is one-time voting authorization?
- How does webcam/biometric verification fit into voting?
- How do I select a candidate?
- How is a vote submitted?
- Can a user vote more than once?

Results and reports

- How are votes counted?
- How are results generated?
- What reports are available?
- What is voter participation reporting?

AI assistant

- How does the AI chatbot work?
- What is RAG?
- What knowledge does the assistant use?
- Can the assistant recommend candidates?
- Can the assistant cast a vote?
- Can the assistant show private voter data?

17. Source Alignment and Interpretation Notes

The first-review PPT is the primary source for the project definition summarized here. It explicitly specifies the project name, abstract, module names, module-level functions, technology stack, hardware/software requirements, and authentication workflow.

Area	PPT support
Project definition	Secure, web-based digital voting platform with AI chatbot.
Authentication	Register, Login, Google Auth, Email Verification, OTP Authentication, Session Management, Security.
Election Creation	Election details/type, voter/candidate configuration, verification setup, schedule/rules.
Voter Verification	Eligibility validation, email OTP, webcam/biometric verification, one-time authorization.
Election Management	Voter/candidate management, start/stop, lock after start, monitoring/audit logs.
Voting	Authorization, candidate selection/confirmation, one-time submission, confidential storage.
Results	Automatic counting, result generation/publishing, participation and candidate/election reports.
AI Chatbot	Election Q&A, voter/creator assistance, context-aware support, knowledge-base/unanswered management.

Interpretation note

Some workflow connections in this document are presented as a coherent end-to-end system flow because the PPT lists the individual module functions rather than a single detailed sequence diagram. Exact implementation details should follow the deployed DigiVote backend/frontend when they differ from this conceptual workflow.

18. Final Summary

DigiVote is an integrated secure digital voting platform with a layered authentication system, election creation and management controls, election-specific voter verification, one-time voting authorization, confidential vote handling, automatic result/report generation, and an AI assistant grounded in project knowledge. The AI assistant should explain these functions accurately, use authorized election context when available, remain neutral, and never bypass or replace the secure voting workflow.